



2025/2026

Lab classes (3)

1 Image processing – Basic manipulations

1.1 Mirror image, Negative image

To mirror an image, just read the horizontal component of the image matrix from end to start

Arithmetic operations can be applied to all pixels – see example of creating a negative image

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("images/guterres-1.jpg")

# Mirror the image
height, width = img.shape[0:2]

# img[lines, cols, color_component]
#      ":" means all (all lines, all columns, all color components -
#      B, G e R)
# img[start:end:step,start:end:step]- just a ":" means all
mirror_img = img[:,width:0:-1]

plt.subplot(2, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title('original')
plt.axis('off')

plt.subplot(2, 2, 2)
plt.imshow(cv2.cvtColor(mirror_img, cv2.COLOR_BGR2RGB))
plt.title('mirror_img')
plt.axis('off')

# Grayscale image
gray_img = cv2.cvtColor(img,
cv2.COLOR_BGR2GRAY)

# Negative image
neg_img = 255 - gray_img

plt.subplot(2, 2, 3)
plt.imshow(gray_img, 'gray')
plt.title('gray level')
plt.axis('off')

plt.subplot(2, 2, 4)
plt.imshow(neg_img, 'gray')
plt.title('negative image')
plt.axis('off')
```



1.2 Visualization of line intensity profile

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

row1 = 350 # line to plot

img = cv2.imread("images/i02.jpg")
gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) # grayscale image

# draw line on the image
width = img.shape[1]
thickness = 2
start_point = (0, row1)
end_point = (width, row1)
color = (128, 128, 128) # Color in BGR
img = cv2.line(img, start_point, end_point, color, thickness)

plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2GRAY), "gray")
plt.title('original image')
plt.show()

### Plot profile of line number "row1"
xaxis = np.arange(0, width, 1)
yaxis = gray_img[row1]

plt.plot(xaxis, yaxis)
plt.title('grayscale line profiles')

plt.show()
```

